P&ID index and line descriptions

Document number: DOC-PID-001 · Revision B · Issued 2025-11-20 · Owner: Engineering documentation

This index describes in text what each piping and instrumentation diagram (P&ID) of the demonstration refinery shows: the main equipment, the process flow between them, and the instruments and control valves on each item. The drawings themselves are PID--.

U-100 Crude distillation unit

Sheet PID-U-100-01: Feed

- **TK-1001 Crude charge tank** (tank; capacity: 20000 m^3, product: Crude oil)
- **P-1001 Charge pump** (pump; pump type: centrifugal, rated flow: 520 m^3/h, power: 450 kW)
  - FT-1001 Feed flow transmitter: flow transmitter, range 0 to 200 m^3/h
  - FV-1001 Feed flow valve: control valve, equal percentage, pneumatic actuator
  - FT-1007 Charge flow transmitter: flow transmitter, range 0 to 200 m^3/h
  - FV-1007 Charge flow valve: control valve, equal percentage, pneumatic actuator
  - FT-1112 Production flow transmitter: flow transmitter, range 0 to 200 m^3/h
  - FV-1112 Production flow valve: control valve, equal percentage, pneumatic actuator

Sheet PID-U-100-02: Heat exchange

- **E-1009 Feed/product exchanger** (heat exchanger; exchanger type: shell and tube, duty: 18.5 MW, area: 640 m^2)
  - TT-1009 Outlet temperature transmitter: temperature transmitter, range 0 to 450 °C
  - TV-1009 Outlet temperature valve: control valve, equal percentage, pneumatic actuator
  - TT-1111 Product temperature transmitter: temperature transmitter, range 0 to 450 °C
  - TV-1111 Product temperature valve: control valve, equal percentage, pneumatic actuator

Sheet PID-U-100-03: Furnaces

- **F-1104 Atmospheric furnace** (furnace; duty: 42 MW, burners: 12, fuel: fuel gas)
  - TT-1104 Furnace temperature transmitter: temperature transmitter, range 0 to 450 °C
  - TV-1104 Furnace temperature valve: control valve, equal percentage, pneumatic actuator
  - TT-1209 Furnace outlet temperature transmitter: temperature transmitter, range 0 to 450 °C

  - TV-1209 Furnace outlet temperature valve: control valve, equal percentage, pneumatic actuator
  - FT-1210 Fuel gas flow transmitter: flow transmitter, range 0 to 200 m³/h
  - FV-1210 Fuel gas flow valve: control valve, equal percentage, pneumatic actuator

Sheet PID-U-100-04: Distillation

- **C-1101 Atmospheric column** (column; trays: 48, diameter: 6.2 m, height: 52 m)
  - FT-1101 Column charge flow transmitter: flow transmitter, range 0 to 200 m³/h
  - FV-1101 Column charge flow valve: control valve, equal percentage, pneumatic actuator
  - TT-1213 Column top temperature transmitter: temperature transmitter, range 0 to 450 °C
  - TV-1213 Column top temperature valve: control valve, equal percentage, pneumatic actuator
  - FT-1214 Side draw flow transmitter: flow transmitter, range 0 to 200 m³/h
  - FV-1214 Side draw flow valve: control valve, equal percentage, pneumatic actuator
- **D-1106 Overhead accumulator** (vessel; volume: 85 m³, design pressure: 3.5 bar g, design temperature: 120 °C)
  - LT-1106 Accumulator level transmitter: level transmitter, range 0 to 100 %
  - LV-1106 Accumulator level valve: control valve, linear, pneumatic actuator
  - LT-1113 Reflux drum level transmitter: level transmitter, range 0 to 100 %
  - LV-1113 Reflux drum level valve: control valve, linear, pneumatic actuator
  - LT-1211 Flash drum level transmitter: level transmitter, range 0 to 100 %
  - LV-1211 Flash drum level valve: control valve, linear, pneumatic actuator
- **P-1102 Reflux pump** (pump; pump type: centrifugal, rated flow: 310 m³/h, power: 160 kW)
  - FT-1102 Reflux flow transmitter: flow transmitter, range 0 to 200 m³/h
  - FV-1102 Reflux flow valve: control valve, equal percentage, pneumatic actuator

Main process flow on these sheets: TK-1001 Crude charge tank → P-1001 Charge pump; P-1001 Charge pump → E-1009 Feed/product exchanger; E-1009 Feed/product exchanger → F-1104 Atmospheric furnace; F-1104 Atmospheric furnace → C-1101 Atmospheric column.

U-200 Hydrotreating unit

Sheet PID-U-200-01: Reaction

- **R-1005 Reactor** (vessel; volume: 120 m³, design pressure: 95 bar g, design temperature: 420 °C)
  - TT-1005 Reactor temperature transmitter: temperature transmitter, range 0 to 450 °C
  - TV-1005 Reactor temperature valve: control valve, equal percentage, pneumatic actuator

Sheet PID-U-200-02: Recycle

- **P-1003 Recycle pump** (pump; pump type: centrifugal, rated flow: 140 m^3/h, power: 75 kW)
  - FT-1003 Recycle flow transmitter: flow transmitter, range 0 to 200 m^3/h
  - FV-1003 Recycle flow valve: control valve, equal percentage, pneumatic actuator

Sheet PID-U-200-03: Separation

- **V-1004 Product separator** (vessel; volume: 60 m^3, design pressure: 12 bar g, design temperature: 180 °C)
  - PT-1002 Vessel pressure transmitter: pressure transmitter, range 0 to 100 bar g
  - PV-1002 Vessel pressure valve: control valve, equal percentage, pneumatic actuator
  - LT-1004 Separator level transmitter: level transmitter, range 0 to 100 %
  - LV-1004 Separator level valve: control valve, linear, pneumatic actuator
  - LT-1010 Interface level transmitter: level transmitter, range 0 to 100 %
  - LV-1010 Interface level valve: control valve, linear, pneumatic actuator
- **V-1105 Surge drum** (vessel; volume: 45 m^3, design pressure: 8 bar g, design temperature: 150 °C)
  - LT-1105 Surge drum level transmitter: level transmitter, range 0 to 100 %
  - LV-1105 Surge drum level valve: control valve, linear, pneumatic actuator

Sheet PID-U-200-04: Compression

- **K-1103 Recycle gas compressor** (compressor; compressor type: centrifugal, power: 2800 kW)
  - PT-1103 Suction pressure transmitter: pressure transmitter, range 0 to 100 bar g
  - PV-1103 Suction pressure valve: control valve, equal percentage, pneumatic actuator
  - PT-1207 Compressor discharge pressure transmitter: pressure transmitter, range 0 to 100 bar g
  - PV-1207 Compressor discharge pressure valve: control valve, equal percentage, pneumatic actuator
- **V-1208 Compressor suction drum** (vessel; volume: 18 m^3, design pressure: 40 bar g, design temperature: 80 °C)
  - LT-1208 Suction drum level transmitter: level transmitter, range 0 to 100 %
  - LV-1208 Suction drum level valve: control valve, linear, pneumatic actuator

Sheet PID-U-200-05: Hydrotreating

- **R-1203 Hydrotreating reactor** (vessel; volume: 95 m^3, design pressure: 110 bar g, design temperature: 430 °C)
  - FT-1201 Hydrotreater charge flow transmitter: flow transmitter, range 0 to 200 m^3/h
  - FV-1201 Hydrotreater charge flow valve: control valve, equal percentage, pneumatic actuator

  - PT-1202 Reactor inlet pressure transmitter: pressure transmitter, range 0 to 100 bar g
  - PV-1202 Reactor inlet pressure valve: control valve, equal percentage, pneumatic actuator
  - TT-1203 Catalyst bed temperature transmitter: temperature transmitter, range 0 to 450 °C
  - TV-1203 Catalyst bed temperature valve: control valve, equal percentage, pneumatic actuator
  - FT-1206 Quench flow transmitter: flow transmitter, range 0 to 200 m^3/h
  - FV-1206 Quench flow valve: control valve, equal percentage, pneumatic actuator
- **V-1204 High-pressure separator** (vessel; volume: 70 m^3, design pressure: 105 bar g, design temperature: 90 °C)
  - LT-1204 High-pressure separator level transmitter: level transmitter, range 0 to 100 %
  - LV-1204 High-pressure separator level valve: control valve, linear, pneumatic actuator
  - AT-1205 Hydrogen purity transmitter: composition transmitter, range 0 to 100 %
  - AV-1205 Hydrogen purity valve: control valve, linear, pneumatic actuator

Sheet PID-U-200-06: Quality

- **AT-1006 Product composition transmitter** (instrument; measured variable: composition, range min: 0, range max: 100, range unit: %)
- **AV-1006 Product composition valve** (control valve; characteristic: linear, actuator: pneumatic)
- **AT-1108 Stream composition transmitter** (instrument; measured variable: composition, range min: 0, range max: 100, range unit: %)
- **AV-1108 Stream composition valve** (control valve; characteristic: linear, actuator: pneumatic)
- **AT-1215 Sulfur content in product transmitter** (instrument; measured variable: composition, range min: 0, range max: 100, range unit: %)
- **AV-1215 Sulfur content in product valve** (control valve; characteristic: linear, actuator: pneumatic)

Main process flow on these sheets: P-1003 Recycle pump → R-1005 Reactor; V-1204 High-pressure separator → K-1103 Recycle gas compressor.

U-300 Treating and sulfur recovery

Sheet PID-U-300-01: Treating

- **C-1216 Amine absorber** (column; trays: 20, diameter: 1.8 m, height: 24 m)
  - LT-1107 Interface level (treating) transmitter: level transmitter, range 0 to 100 %
  - LV-1107 Interface level (treating) valve: control valve, linear, pneumatic actuator
  - TT-1114 Stabilizer temperature transmitter: temperature transmitter, range 0 to 450 °C

  - TV-1114 Stabilizer temperature valve: control valve, equal percentage, pneumatic actuator
  - LT-1216 Amine absorber level transmitter: level transmitter, range 0 to 100 %
  - LV-1216 Amine absorber level valve: control valve, linear, pneumatic actuator

Sheet PID-U-300-02: Sulfur recovery

- **R-1226 Claus reactor** (vessel; volume: 35 m^3, design pressure: 1.5 bar g, design temperature: 350 °C)
  - TT-1226 Sulfur reactor temperature transmitter: temperature transmitter, range 0 to 450 °C
  - TV-1226 Sulfur reactor temperature valve: control valve, equal percentage, pneumatic actuator

U-900 Utilities and offsites

Sheet PID-U-900-01: Utilities

- **P-1219 Cooling water pump** (pump; pump type: centrifugal, rated flow: 1800 m^3/h, power: 630 kW)
  - FT-1219 Cooling water flow transmitter: flow transmitter, range 0 to 200 m^3/h
  - FV-1219 Cooling water flow valve: control valve, equal percentage, pneumatic actuator
  - TT-1220 Water return temperature transmitter: temperature transmitter, range 0 to 450 °C
  - TV-1220 Water return temperature valve: control valve, equal percentage, pneumatic actuator
- **V-1218 Condensate drum** (vessel; volume: 25 m^3, design pressure: 6 bar g, design temperature: 165 °C)
  - FT-1109 Steam flow transmitter: flow transmitter, range 0 to 200 m^3/h
  - FV-1109 Steam flow valve: control valve, equal percentage, pneumatic actuator
  - PT-1217 Medium-pressure steam pressure transmitter: pressure transmitter, range 0 to 100 bar g
  - PV-1217 Medium-pressure steam pressure valve: control valve, equal percentage, pneumatic actuator
  - LT-1218 Condensate drum level transmitter: level transmitter, range 0 to 100 %
  - LV-1218 Condensate drum level valve: control valve, linear, pneumatic actuator
- **PT-1008 Header pressure transmitter** (instrument; measured variable: pressure, range min: 0, range max: 20, range unit: bar g)
- **PV-1008 Header pressure valve** (control valve; characteristic: equal percentage, actuator: pneumatic)
- **PT-1110 Network pressure transmitter** (instrument; measured variable: pressure, range min: 0, range max: 20, range unit: bar g)
- **PV-1110 Network pressure valve** (control valve; characteristic: equal percentage, actuator: pneumatic)

- **PT-1212 Flare header pressure transmitter** (instrument; measured variable: pressure, range min: 0, range max: 20, range unit: bar g)
- **PV-1212 Flare header pressure valve** (control valve; characteristic: equal percentage, actuator: pneumatic)
- **PT-1221 Nitrogen network pressure transmitter** (instrument; measured variable: pressure, range min: 0, range max: 20, range unit: bar g)
- **PV-1221 Nitrogen network pressure valve** (control valve; characteristic: equal percentage, actuator: pneumatic)

Sheet PID-U-900-02: Water treatment

- **TK-1223 Equalization tank** (tank; capacity: 3500 m^3, product: Process water)
  - FT-1222 Effluent flow to treatment transmitter: flow transmitter, range 0 to 200 m^3/h
  - FV-1222 Effluent flow to treatment valve: control valve, equal percentage, pneumatic actuator
  - LT-1223 Equalization tank level transmitter: level transmitter, range 0 to 100 %
  - LV-1223 Equalization tank level valve: control valve, linear, pneumatic actuator
  - AT-1224 Treated effluent pH transmitter: composition transmitter, range 0 to 100 % of analyzer span (0 to 14 pH)
  - AV-1224 Treated effluent pH valve: control valve, linear, pneumatic actuator
- **TK-1225 Slop tank** (tank; capacity: 1200 m^3, product: Slop oil)
  - LT-1225 Slop tank level transmitter: level transmitter, range 0 to 100 %
  - LV-1225 Slop tank level valve: control valve, linear, pneumatic actuator